Lecture 10

Thursday, 9 February 2023 1:32 PM

Electrochem Rxn Equilibrium

- Phase Equilibrium: chemical species in different phases are in equilibrium if their chem. potentials are equal

$$\mu_j^{\alpha} = \mu_j^{\beta}$$

Eg: O_2 dissolved in water: $\mu_{O_2}^{aq} = \mu_{O_2}^{g}$ → Henry's Law $P_{O_2} = K_H [O_2]$ → Chemical potential equivalence b/w gas phase & liquid solubility of a species

- Chemical Rxn Equilibrium: A chem rxn is in equil when chem. potential of reactants & products are equal

$$A + B \rightleftharpoons C$$

$$\mu_A + \mu_B = \mu_C$$

Conc. @ standard conditions

↳ For ideal solutions (activity coeff = 1): $\mu_j = \mu_j^{\circ} + RT \ln \frac{c_j}{c_j^{\circ}}$

(μ_j°: chem. potential @ std. conditions)

Plugging this in for our reaction:

$$\frac{C_C^{\circ}}{C_A^{\circ} C_B^{\circ}} \exp\left[\frac{\mu_A^{\circ} + \mu_B^{\circ} - \mu_C^{\circ}}{RT}\right] = \frac{C_C}{C_A C_B}$$

$$K = \frac{C_C}{C_A C_B}$$

For an echem rxn, we can write: $\sum s_j \mu_j \rightleftharpoons n e^-$

@ equil,

$$\sum s_j \mu_j = n \mu_{e^-}$$

→ This is what we measure via voltmeter (electrochem potential of e^-)

Ex: Consider the full cell rxn:

$$\underset{\alpha}{Pt} \mid \underset{\beta}{H_2(g)} \mid \underset{\gamma_1}{H_2SO_4(aq)} \parallel \underset{\gamma_2}{CuSO_4(aq)} \mid \underset{s}{Cu(s)} \mid \underset{\alpha'}{Pt(s)}$$